Zain Rizvi

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Weston, FL | Dual US / Canadian Citizen

Education

Northeastern University Sep. 2025 – May 2029
Bachelor of Science in Computer Engineering and Computer Science — GPA: 3.83 *Boston, MA*

- **Relevant Coursework:** Data Structures & Algorithms, Differential Equations & Linear Algebra, Object-Oriented Design, Databases, Discrete Structures, Physics 1

Cypress Bay High School Aug. 2023 – May 2025
GPA: 3.8/4.0 — 11 AP Courses (all scores 4 or 5) *Weston, FL*

Experience

Machine Learning Research Intern Jun. 2026 – Present
Florida International University *Miami, FL*

- Contributing to a federated learning project on energy grid data in partnership with FPL (Florida Power & Light), under Dr. Arif Sarwat
- Building machine learning pipelines in Python and scikit-learn, including data partitioning, model training, and evaluation
- Prototyped a federated setup on a digits benchmark dataset where models train locally across distributed clients and aggregate weights via FedAvg without centralizing raw data

Project Lead, Research Practicum Jan. 2026 – May 2026
Northeastern University, Oak Ridge National Laboratory *Boston, MA*

- Led 4 undergraduates and worked with graduate students to design, build, and deploy the application
- Built and deployed a research web application (TypeScript, React, Next.js on Vercel) at the request of ORNL PI Dr. Rajeev Kumar, providing browser-based access to Polymer Chemistry-Informed Neural Network (PCINN) predictions and Molecular Weight Distribution data
- Developed a parameter sweep tool that varies a single input across a range and updates all outputs in real time, replacing manual one-at-a-time prediction runs, plus batch prediction from uploaded files
- Designed an Arrhenius analysis feature from scratch enabling scientists to validate experimental results against model predictions across temperature ranges
- Implemented an in-app model training interface with configurable hyperparameters and live progress, moving training from a separate offline step into the application

Summer Scholar, Clinical Rotations Jun. 2024 – Aug. 2024
Cleveland Clinic *Weston, FL*

- Participated in clinical rotations across specialties, including internal medicine, surgery, and cardiology
- Gained hands-on experience in patient care and clinical decision-making under physician mentorship

Projects

Novi | *Swift, SwiftUI, Firebase, iOS Development* 2025

- Built an iOS roommate-matching app in Swift and SwiftUI with a swipe-based matching interface
- Designed user profiles with preferences for living habits, budget, and location to optimize compatibility

Miniature Traffic Control System | *Arduino, C++, Electronics* 2024

- Designed and built a functional miniature traffic control system using Arduino microcontroller technology
- Presented project at Science & Technology Conference, demonstrating embedded systems concepts

Technical Skills

Languages: Python, C++, Java, Swift, TypeScript, HTML/CSS, JavaScript, SQL
Technologies: React, Next.js, Arduino, Neural Networks, Machine Learning, Git, Linux, SwiftUI, Firebase, Databricks
Tools: VS Code, Vercel, Google Cloud Platform, Xcode

Honors & Achievements

Dean's List (Fall 2025, Spring 2026) · AP Scholar with Distinction (July 2025) · Hassan Wadi Youth Academy